

Documenting our Input output code

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1 Checking setup

```
import sys
print("hello from IPython session")
print("executable:", sys.executable)
```

2 National IO accounts

3 National A Program

We document building the national A table

```
import numpy as np
import pandas as pd
```

Now we read a spreadsheet and get the sheet names

```
data_pth = "../data/raw/input_output/"
fname = f"{data_pth}IOUse_After_Redefinitions_PRO_Detail.xlsx"
dfs = pd.read_excel(fname, sheet_name=None)
```

Pull out the use table

```
use = dfs['2017']
```

Put the flows into a numpy array

```
flows = use.values[5:, 2:]
```

The commodity? labels are in the first column:

```
labels = use.values[5:, 0]
```

And the industry labels:

```
columns = use.iloc[4].values[2:]
```

Handle the nans by setting them to 0: flows = use.values[5:, 2:] The commodity? labels are in the first column:

```
use_df = pd.DataFrame(data=flows, columns=columns, index=labels).fillna(0)
```

Build a commodity dataframe

```
commodity_names = use.values[5:,1]
commodity_codes = use.values[5:,0]
commodity_df = pd.DataFrame(data=use.values[5:,0:2], columns=['Code', 'Description'])
```

And an industry dataframe

```
industry_df = pd.DataFrame(use.values[3:5,3:].T, columns=['Description', 'Code'])
```

Now the Make table

```
make_dfs = dfs = pd.read_excel("../data/raw/input_output/IOMake_After_Redefinitions_PR
make = make_dfs['2017']
columns = make.iloc[4].values[2:]
flows = make.values[5:,2:]
labels = make.values[5:,0]
make_df = pd.DataFrame(data=flows, columns=columns, index=labels).fillna(0)
```

Normalizing the tables

```
U = use_df.values[:400,:]
X = use_df.values.sum(axis=0)
Xd = np.diag(1/(X+(X==0)))
B = U @ Xd
Q = make_df.values.sum(axis=0)
Qd = np.diag(1/(Q + (Q==0)))
D = make_df.values @ Qd
```

A matrix

```
A = D[:-1, :400] @ B
print(A.shape)
print(A[0:5,1])
A = A[:, :402]
print(A.shape)
L = np.linalg.inv(np.eye(402) - A)
print(L.sum(axis=0))
```